

Safe reduction of conservatism by combining machine learning and physics-based models

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Complex engineering systems are often designed using advanced physics-based models to understand the critical limitations of the system before building, installing, or operating it. During design, it is prudent to apply a set of conservative assumptions. Some are needed to handle stochastic variability in what the system will experience during its lifetime, i.e. *aleatory uncertainty*, while others are made because we lack knowledge, i.e. *epistemic uncertainty*. We need for example to make conservative assumptions related to relevant weather scenarios to ensure that an offshore structure can withstand the induced loads it may experience during its lifetime. Some aspects are unknown during the design phase, but during operation, we can gather evidence and update our knowledge to reduce the applied conservatism without compromising the acceptable risk level of the system. This paper describes a method to optimize the use of both data and physics-based models in high-risk systems through: the rigorous distinction between assumptions that include necessary conservatism and assumptions where conservatism can be reduced by knowledge gained through operational experience; probabilistic assessment of model discrepancy between physics-based and data-driven models; and iterative Bayesian update of model assumptions. It showcases how to reduce the model discrepancy between physics-based models and experience-based data-driven models (machine learning) without compromising safety.

Keywords: probabilistic machine learning, Gaussian process regression, high-risk decisions, operating limits, uncertainty quantification, extreme value analysis, Bayesian statistics, weather-induced loads.

1. Introduction

Current methods seem to be conservative compared to what we observe from sensor data while models based on sensor data do not capture safety-critical behavior not yet experienced. Neither models give optimal decision support and we are left with conservatism to offset the uncertainty.

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Modern machine learning (ML) methods and self-learning systems have proven to be very effective to solve hard engineering problems across a wide set of domains. However, in industries where the value of what is produced is high and where the negative impact of a disrupted service is significant, or where operational deviations could be dangerous, it is of utter importance that control strategies and critical decisions are based on the best possible scientific understanding as well as all relevant experience. Current engineering practices

may not reflect this if it is based on either physics-based models (causal knowledge) or data-driven models (empirical experience) alone, and not the combination of the two.

Physics-based and data-driven methods have the same objective; To make sure that the control of-, and decisions related to, the systems under study are optimal and safe. However, these approaches are fundamentally different, with different strengths and weaknesses.

Physics-based models seek to simulate key mechanisms behind a system's behavior and can be used to explore critical scenarios outside the current experience base. Yet they are inevitably imperfect idealizations of reality, and often computationally expensive. Data-driven models offer fast answers and insights about the non-idealized behavior of complex systems from experience, but introduce uncertainties from observations and a limited experience base. These limitations narrow



Fig. 1. The Rowan Viking jack-up drilling rig is positioned immediately next to the bottom-fixed Edvard Grieg platform.

the scope where data-driven models can be used as well as the accuracy and timely response of physics-based models.

1.1. Problem description

Critical operational decision criteria are often built on conservative assumptions to offset uncertainties that might lead to potential negative consequences. The conservatism is included to handle both aleatory (variability) and epistemic (knowledge-related) uncertainties.

How can operators of critical and high-consequence systems safely reduce overly conservative assumptions made in the design to benefit from acquired knowledge gained through operational experience, while still ensuring that the system is robust enough to survive rare (and not yet experienced) events?

This paper discusses and demonstrates a process of how to approach the overarching problem of optimizing critical decisions for rare events in an operational setting without compromising the required risk level. It exemplifies this by looking at a problem where we have access to a simulator that represents a physical process in addition to measurements of the response of this process.

1.2. Proposed process

To address the overarching problem of optimizing critical decisions in an operational setting, this paper suggests an iterative Bayesian approach to solve the following sub-problems of safe model

calibration:

- (i) Establish consistency of feature (input) space and responses (output) across both simulator and observations.
- (ii) Distinguish between necessary and reducible conservatism.
 - (i) Identify aleatory conservative aspects related to the robustness of future variability.
 - (ii) Identify epistemic conservative aspects related to lack-of-knowledge.
- (iii) Assess model discrepancy
 - (i) Compare data-driven and physics-based models across the entire relevant input space (e.g. through surrogate models as described in more detail in section 3.1).
 - (ii) Assess potential improvements for both the data-driven and the physics-based models based on identified discrepancies and model understanding.
- (iv) Model calibration: Update the models and relevant assumptions based on new information/knowledge
 - (i) Update/transform the data-driven model to reflect the response(s) considered in the physics-based model.
 - (ii) Update relevant assumptions in the physics-based model based on knowledge gained from the observed data.
- (v) Assess confidence level in decision criteria

- (vi) Assess resource needs and potential benefit gain.
 - (i) If the resource use and benefits gained are viable – *Reiterate the steps iv to vi.*
 - (ii) If the resource use and benefits gain are not viable – *Stop* (the current level of conservatism cannot practically and safely be further reduced).

2. Use case: Weather-induced relative motion of two offshore platforms

Lundin Energy Norway operates several expensive and highly complex production systems for which they have to make high-consequence decisions daily. One such decision is whether or not to continue drilling operations through upcoming bad weather. During drilling operations, a jack-up drilling rig is positioned immediately next to the bottom-fixed Edvard Grieg (EG) platform with its drilling derrick positioned above EG's weather deck as shown in Fig. 1.

During drilling operations, the jack-up rig is connected to the wellhead with a high pressure riser system. Any relative movement between the jack-up rig and the platform will result in forces acting on components in the high pressure riser and wellhead systems. Some of these components are critical, in the sense that if they are subjected to excessive forces above their capacity, they can break and lead to the loss of integrity of the well.

Weather and sea will induce loads on both the jack-up and bottom-fixed platforms, which in turn might lead to relative motion between the two independent platforms. Thus, before the operation starts the operator needs to make sure that any situation that might induce critical relative motions and the related potential failure modes are sufficiently understood and controlled.

The failure mechanisms in this system are related to an over-utilization of bending moment in a critical part of the riser stack-up and the fatigue damage accumulated by cyclic loading of a critical part of the riser stack-up. However, the system's capacity concerning the critical failure properties can be correlated to a maximum relative motion. In this use case, we will look at this relative motion as a proxy for the underlying failure

mechanisms.

In the following, we adopt the critical capacity of the system as relative motions, d , at the location where the high pressure riser is connected to the wellhead, at the cellar deck of the EG platform. We denote the critical capacity d_c and defined the limit state function by:

$$g(d) = d - d_c \quad (1)$$

where $g(d) < 0$ signifies a failure state.

In the design phase, a set of dynamic finite element (FE) (i.e. causal- and physics-based) models have been run to estimate the maximum relative motion the jack-up and bottom-fixed structures might experience for different weather scenarios. In this example the weather scenarios are represented by the following environmental parameters: the significant wave height, H_s , the primary wave peak period, T_p , wind speed, u_w , current speed u_c , and the corresponding directions $\theta_{wave}, \theta_{wind}, \theta_{current}$. Based on the simulated relative motion from several FE model runs, a statistical extreme value (EV) of relative motion has been estimated as a function of different weather scenarios. The approach here is to fit a Weibull distribution to the observed/simulated maxima of a three-hour sea state and estimate the 100-year response as 90% of the corresponding Gumbel distribution (see e.g. Sødahl et al. (2006), DNV (2021) and Norsk Standard (2017) for details).

As offshore operations are complex and have potentially dire consequences of erroneous decisions, a simple and robust decision criterion is desirable. In this use case, the initial FE model results showed that the significant wave height, H_s , is dominating the relative motion response. The operational limit was defined based on when the simulated EV of the relative motion exceeded the critical level. The decision to operate or not can be formulated as:

$$\mathcal{A}(H_s) = \begin{cases} \text{operate} & H_s \leq H_s^{olim} \\ \text{disconnect} & H_s > H_s^{olim} \end{cases} \quad (2)$$

Where $\mathcal{A}$ is the set of actions to operate or not as a function of the H_s . Fig. 2 shows the FE model

results (light blue) and the maximum observed relative motion (dark blue) as a function of sea-state parameters H_s and T_p .

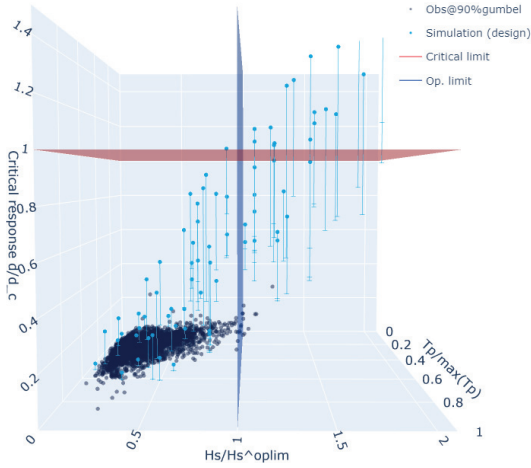


Fig. 2. Critical relative motion and operating limit plotted together with FE model results (light blue) with the static offset (vertical negative error-bar) and observations (dark blue) in normalized H_s, T_p input space.

The critical threshold $d = d_c$ is shown in Fig. 2 as a red plane, and the corresponding operating limit based on where the initial FE results exceed the critical threshold is shown as the dark blue vertical plane in Fig. 2.

After logging the measured relative motion for approximately 500 days of operation over 3 years, the logged maximum relative motion (dark blue dots in Figure 2) shows significantly less relative motion than estimated by the FE model for weather scenarios with increasing H_s . This raises a few important questions:

- Is the *measured* relative motion representing the same *simulated* relative motion from the FE model used to establish the critical limit?
- Has the system experienced statistically relevant extreme events used as the basis for the FE model to ensure that it is robust towards rare, but relevant, events (i.e. related to aleatory uncertainty)?
- Does the FE model contain conservative assumptions that can be reduced based on new knowledge about the system's experienced re-

sponse (i.e. conservatism related to epistemic uncertainty)?

Where the first question relates to step (i) of the proposed process in section 1.2 while the two others relate to step (ii).

2.1. Measurement system

The logged relative motion between the jack-up and the bottom-fixed structures was measured using a laser-based sensor system. However, this system was only able to measure relative dynamic motion between the two rigs. Any static offset would be difficult to capture in an operational setting as it would require consistent and frequent re-calibration of the measurement system.

The original design simulations, which include both static offset and dynamic oscillations around an equilibrium, cannot be compared to the measurements without accounting for the static offset. In Fig. 2 the static part of the simulated relative motion has been subtracted through a principal component analysis Pearson (1901), and this static offset is plotted as the vertical negative error bars.

This explains a substantial part of the difference between the measured and simulated system response, but the observations are still substantially below the dynamic motion of the FE model.

2.2. Weather scenarios

Fig. 3 shows the historical distribution of weather scenarios described by the significant wave height, H_s , and primary wave peak period, T_p , in green. The distribution of observed weather scenarios during the 3 years of operation is plotted as dark blue dots. The FE model runs are plotted in light blue. The critical decision criteria are shown as a dark blue line.

From Figs. 2 and 3 it is also clear that in the three years of observations only a few scenarios have been close to or above the critical decision criteria $H_s \geq H_s^{optim}$. This supports the statement that models based on sensor data do not capture safety-critical behavior not yet experienced, and thus critical decisions based on a purely data-driven model are associated with considerable uncertainty.

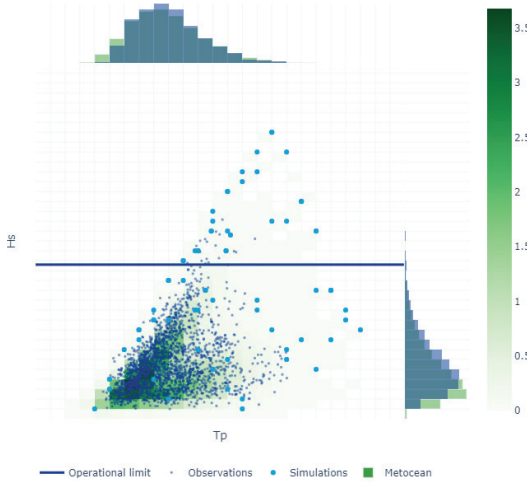


Fig. 3. Historical (green) weather scenarios overlaid with observed (dark blue) and FE model runs (light blue).

However, the observations from the three years of operation contain considerable information about the real response of the system under the observed weather scenarios. This information can be used to calibrate the model so that assumptions made in the design, that potentially are too conservative, can be reduced, or so that erroneous unconservative assumptions can be rectified.

3. Model discrepancy & calibration

To be able to calibrate the model it is important to assess the discrepancy between the simulation model and the real-world observations across the entire input space. As the simulations are generally run for idealized scenarios that are not one-to-one with the scenarios observed in the real world, one approach is to establish two surrogate models, one based on the observed data, and one based on the simulated data, and use these for the comparison.

3.1. Gaussian process surrogate model

We want to adopt a surrogate model as a fast approximation of both the costly simulator and the real-world process. That is, we want to approximate the costly simulation (or real-world process) f with an approximative function $\hat{f}$ such that:

$$\hat{f}(x) \approx f(x), \quad (3)$$

using observations of the form $\mathcal{D} = \{(x_i, y_i)\}$ where $y_i = f(x_i) + \epsilon_i$ is the observed response corresponding to input x_i and the noise terms ϵ_i are assumed independent and identically distributed Gaussian.

A Gaussian process (GP) is a stochastic process where any finite linear combination of the random variables has a multivariate normal distribution. See Rasmussen and Williams (2006) for a detailed introduction to GPs and their use in machine learning. A GP,

$$\hat{f} \sim \mathcal{GP}(\mu(x), K(x, x')) \quad (4)$$

is fully defined by its mean ($\mu(x)$) and covariance ($K(x, x')$) functions and allows prediction of $f(x^*)$ at new observation locations x^* . The predictive posterior $\hat{f}(x^*)|\mathcal{D}$ is still Gaussian with mean and covariance that can be computed analytically Rasmussen and Williams (2006).

We have selected a mean function with the expected value equal to the critical limit. This forces the predictions to approach the critical limit if one does not have data to support that it is less than the critical limit, i.e. that the expected value of the prior $\mu = d_c$. For a physical system with a continuous response, we have selected a Matérn 5/2 covariance (kernel) function which is twice differentiable with an anisotropic lengthscale along each of the input dimensions (see Rasmussen and Williams (2006) for details).

The GPs in this work have been optimized using the python package GPyTorch Gardner et al. (2018) and the optimization routines of the tensor package PyTorch Paszke et al. (2019). The GPs have been optimized using the Adam optimizer with a Marginal log-likelihood loss function.

3.2. Conservative assumptions related to the eigenfrequency

The GP models based on the dynamic maximum response of the FE model results (denoted GP|fe) and the measured observations (denoted GP|obs) have been compared in Fig. 4. These predictions are made for increasing H_s where both $T_p \propto H_s$ and $u_w \propto H_s$ follow a linear relation with H_s between the historically most likely T_p and u_w at the minimum and maximum of the observed H_s

(see Fig. 3). The light blue curve of the predicted response from the GP|fe is significantly higher than the predictions based on the GP|obs. The remaining input has been kept fixed. The directions are aligned, $\theta_{wave} = \theta_{wind} = \theta_{current}$, at one of the dominating directions with the highest observed response as it is assumed to be conservative with all environmental forces acting in the same direction. The current u_c has been set larger than most of the observed currents. Note that the observations plotted include all observations across the $n_x = 7$ -dimensional input space. Fig. 4 shows that the GP|obs matches well the observations below H_s^{optim} and that the GP|fe significantly overpredict the dynamic response close to the H_s^{optim} .

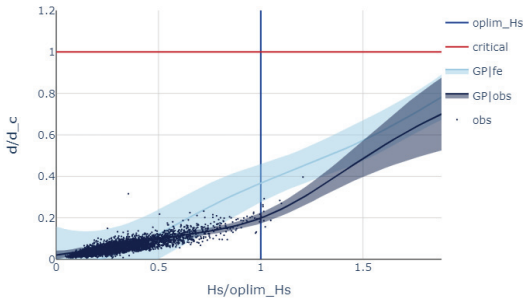


Fig. 4. Dynamic response predicted by GPs at relevant H_s , T_p , u_w combinations based on the dynamic response of original FE model (light blue) and the original observations (dark blue). All observations are in dark blue dots.

One reason for the discrepancy in the dynamic motion may be related to the stiffness of the system. Fig. 5 shows Welch's power spectrum density for real-world observations from 6 selected storms and the corresponding FE model runs. The first peak at approx. 0.8 Hz (or 12.5 s) is the main frequency (or period) of the wave loads. The second peak is the eigenfrequency of the system. The real-world observations (solid lines) have an eigenfrequency of approx. 0.15 Hz which is significantly above the eigenfrequency observed in the FE model of approx. 0.12 Hz. The difference in eigenfrequency means that the system in the FE model is "softer" than the real-world system, and thus experiences larger dynamic oscillations.

The soil in which the jack-up rig is placed is

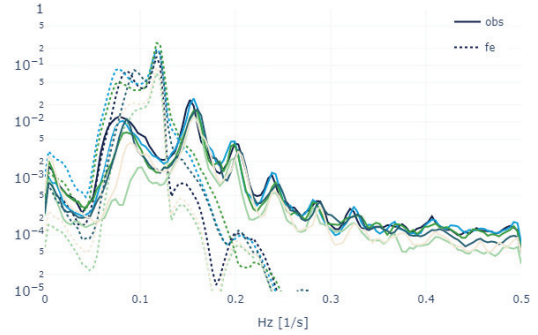


Fig. 5. Welch's power spectrum density of observations (solid) and FE models (dashed) of 6 selected storms.

difficult to measure a priori. Thus, in design, one uses a conservative assumption. However, after the system is set in operation, one can gain empirical evidence that supports a less conservative assumption on e.g. the soil stiffness. Fig. 6 shows how changing the soil stiffness parameter between a lower and upper bound (denoted fe_lb and fe_ub) affects the eigenfrequency of the FE model and how it approaches the real-world eigenfrequency.

Note that this is only one example of a parameter that can be calibrated by comparing real-world data and synthetic data generated by the FE model. It illustrates how a more accurate model can be achieved by calibration of model parameters to fit knowledge gained by empirical experience.

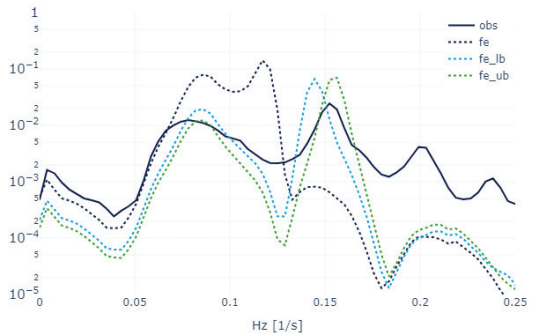


Fig. 6. Welch's power spectrum density of observation (solid) and corresponding FE model with three different soil stiffness parameters (dashed).

3.3. Iterate and update model assumptions

A new GP is trained on the dynamic maximum response of both the observations and the updated FE model. Fig. 7 shows the GP predictions across a large Latin Hypercube Sample (LHS) from the relevant environmental states (see Fig. 3) for the GPs based on the original FE model (light blue) and the updated soil stiffness (red). As can be seen from Fig. 7 the GP predictions based on the original FE model have a larger variation across the sampled input space compared to the GP based on both observations and the updated FE model.

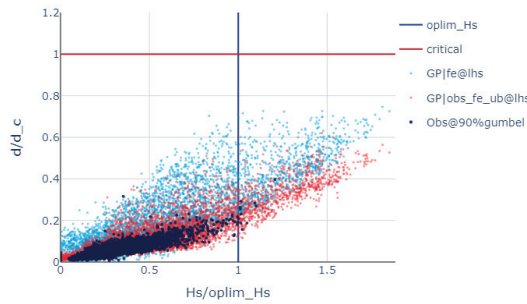


Fig. 7. Dynamic response predicted by GP at LHS based on the original FE model (light blue) and the combination of the updated FE model and observations (red). Original observations (dark blue).

Table 1 shows the Mean absolute error (MAE), the Root mean squared error (RMSE), and the Coefficient of determination (R^2 score) for the GPs based on the FE model, the observations, and the observations and the updated FE model when used to predict the original observations. It also presents these metrics for the subset of observations above the H_s^{oplim} . The GP|obs+fe_ub scores significantly better on these metrics.

A GP has also been trained on the static response of the updated FE model. This enables a combination of the GPs based on both observations and the updated FE model to represent the total displacement. This is plotted in red in Fig. 8 together with the GP based on the total displacement of the original FE model in light blue for comparison along the relevant H_s , T_p , and u_w as done for the dynamic response in Fig.

Table 1. Regression metrics.

Metric	GP fe	GP obs	GP obs+fe_ub
@observations			
RMSE	0.07	0.0091	0.0089
MAE	0.05	0.0073	0.0072
R^2	-8.4	0.84	0.85
@obs $\geq H_s^{oplim}$			
RMSE	0.08	0.037	0.010
MAE	0.07	0.031	0.009
R^2	-3.4	0.14	0.91

4.

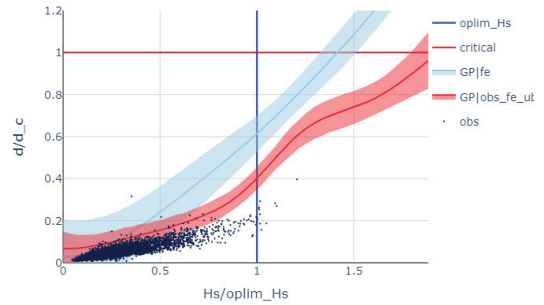


Fig. 8. Total response predicted by GPs at relevant H_s , T_p , u_w combinations based on the dynamic and static response of the original FE model (light blue) and the combination of the updated FE model and observations (red). Original observations (dark blue).

4. Results and conclusion

Fig. 9 shows the predicted response of the LHS considering the entire possible environmental input space. After gaining more operational experience and knowledge about the real-world response, the FE model has been updated, and the prediction model is based on both operational experience and calibrated physics-based modeling for scenarios not yet experienced.

Based on this updated surrogate, the total response at the upper 95% confidence limit does not exceed the critical limit until the H_s is 25% larger than the original operational limit. From the historical weather scenarios illustrated in Fig. 3, the original H_s^{oplim} corresponds to weather scenarios that typically would be experienced once

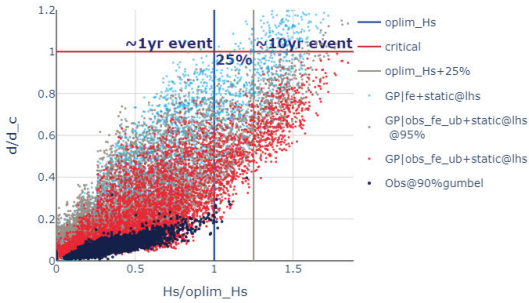


Fig. 9. Total response predicted by GPs at LHS based on the dynamic and static response of the original FE model (light blue) and the combination of the updated FE model and observations (red). The upper 95% confidence bound on the latter prediction is plotted in gray. Original dynamic observations in dark blue.

every year (i.e. a 1-year event). If one can increase the operating limit based on the updated and calibrated model, a 25% increase in acceptable H_s scenarios corresponds to weather events that are only experienced once every 10 years. This increases the operational window significantly without compromising on the conservatism needed to be robust against future extreme events.

This paper has illustrated a structured approach to reducing unnecessary conservatism by incorporating operational experience, calibrating physics-based models, and combining the knowledge from both experience and calibrated physics-based models safely. The example showcases the benefit of the approach by increasing the operational limits significantly without compromising on safety. It is a robust and simple approach to combining machine-learning and physics-based models to benefit from the strengths of both domains.

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